

C Additional Fact Retrieval Results

Relation	Manual Prompt (LAMA)	#train	LAMA	LPAQA	AUTOPROMPT
P1001	[X] is a legal term in [Y]	1000	70.47	72.75	82.45
P101	[X] works in the field of [Y]	864	9.91	5.32	12.79
P103	The native language of [X] is [Y]	1000	72.16	72.16	82.09
P106	[X] is a [Y] by profession	1000	0.63	0.0	14.72
P108	[X] works for [Y]	376	6.79	5.74	8.62
P127	[X] is owned by [Y]	548	34.79	32.46	35.95
P1303	[X] plays [Y]	1000	7.59	18.02	15.38
P131	[X] is located in [Y]	1000	23.27	22.81	37.46
P136	[X] plays [Y] music	1000	0.75	16.76	55.42
P1376	[X] is the capital of [Y]	310	73.93	59.83	40.17
P138	[X] is named after [Y]	856	61.55	59.69	66.05
P140	[X] is affiliated with the [Y] religion	445	0.63	59.83	75.26
P1412	[X] used to communicate in [Y]	1000	65.02	64.71	71.21
P159	The headquarter of [X] is in [Y]	1000	32.37	35.57	35.47
P17	[X] is located in [Y]	1000	31.29	35.48	52.15
P176	[X] is produced by [Y]	1000	85.64	81.67	87.78
P178	[X] is developed by [Y]	560	62.84	59.12	66.72
P19	[X] was born in [Y]	1000	21.08	20.87	19.92
P190	[X] and [Y] are twin cities	895	2.41	1.91	2.31
P20	[X] died in [Y]	1000	27.91	27.91	31.16
P264	[X] is represented by music label [Y]	1000	9.56	10.26	43.82
P27	[X] is [Y] citizen	1000	0.0	41.51	46.69
P276	[X] is located in [Y]	1000	41.5	41.5	44.11
P279	[X] is a subclass of [Y]	1000	30.74	14.75	54.93
P30	[X] is located in [Y]	1000	25.44	18.56	70.36
P31	[X] is a [Y]	1000	36.66	36.66	51.95
P36	The capital of [X] is [Y]	1000	62.16	62.16	60.6
P361	[X] is part of [Y]	1000	23.61	31.44	17.7
P364	The original language of [X] is [Y]	1000	44.51	43.93	48.48
P37	The official language of [X] is [Y]	311	54.55	56.83	62.63
P39	[X] has the position of [Y]	1000	7.96	16.14	30.72
P407	[X] was written in [Y]	1000	59.18	65.22	68.42
P413	[X] plays in [Y] position	1000	0.53	23.74	41.7
P449	[X] was originally aired on [Y]	1000	20.89	9.08	34.39
P463	[X] is a member of [Y]	679	67.11	57.33	54.22
P47	[X] shares border with [Y]	1000	13.67	13.34	19.52
P495	[X] was created in [Y]	1000	16.5	32.23	36.63
P527	[X] consists of [Y]	1000	11.07	10.55	25.61
P530	[X] maintains diplomatic relations with [Y]	927	2.81	3.92	3.11
P740	[X] was founded in [Y]	1000	7.59	13.68	13.89
P937	[X] used to work in [Y]	1000	29.77	39.1	38.36

Table 6: A breakdown of all relations for fact retrieval on the original dataset from [Petroni et al. \(2019\)](#). We compare P@1 of prompts generated by LAMA, LPAQA, and our approach using five prompt tokens.

Relation	Method	Prompt	P@1
P101	Manual	[X] works in the field of [Y]	11.52
	AUTOPROMPT BERT	[X] probability earliest fame totaled studying [Y]	15.01
	AUTOPROMPT RoBERTa	[X] 1830 dissertation applying mathsucci [Y]	0.17
P103	Manual	The native language of [X] is [Y]	74.54
	AUTOPROMPT BERT	[X]PA communerug speaks proper [Y]	84.87
	AUTOPROMPT RoBERTa	[X]neau optionally fluent!?!traditional [Y]	81.61
P106	Manual	[X] is a [Y] by profession	0.73
	AUTOPROMPT BERT	[X] supporters studied politicians musician turned [Y]	15.83
	AUTOPROMPT RoBERTa	[X] (), astronomers businessman·former [Y]	19.24
P127	Manual	[X] is owned by [Y]	36.67
	AUTOPROMPT BERT	[X] is hindwings mainline architecture within [Y]	47.01
	AUTOPROMPT RoBERTa	[X] picThom unwillingness officially governs [Y]	39.58
P1303	Manual	[X] plays [Y]	18.91
	AUTOPROMPT BERT	[X] playingdrum concerttoative electric [Y]	42.69
	AUTOPROMPT RoBERTa	[X]Trump learned soloKeefe classical [Y]	44.44
P136	Manual	[X] plays [Y] music	0.7
	AUTOPROMPT BERT	[X] freaking genre orchestra fiction acid [Y]	59.95
	AUTOPROMPT RoBERTa	[X] blends postwar hostage drama sax [Y]	52.97
P1376	Manual	[X] is the capital of [Y]	81.11
	AUTOPROMPT BERT	[X] boasts native territory traditionally called [Y]	63.33
	AUTOPROMPT RoBERTa	[X] limestone depositedati boroughDepending [Y]	28.33
P178	Manual	[X] is developed by [Y]	62.76
	AUTOPROMPT BERT	[X] is memory arcade branding by [Y]	64.45
	AUTOPROMPT RoBERTa	[X] 1987 floppy simulator users sued [Y]	69.56
P20	Manual	[X] died in [Y]	32.07
	AUTOPROMPT BERT	[X] reorganizationotype photographic studio in [Y]	33.53
	AUTOPROMPT RoBERTa	[X].. enigmatic twentieth nowadays near [Y]	31.33
P27	Manual	[X] is [Y] citizen	0.0
	AUTOPROMPT BERT	[X] m³ badminton pieces internationally representing [Y]	46.13
	AUTOPROMPT RoBERTa	[X] offic organise forests statutes northwestern [Y]	42.07
P276	Manual	[X] is located in [Y]	43.73
	AUTOPROMPT BERT	[X] consists kilograms centred neighborhoods in [Y]	44.64
	AUTOPROMPT RoBERTa	[X] manoeuv constructs whistleblowers hills near [Y]	37.47
P279	Manual	[X] is a subclass of [Y]	31.04
	AUTOPROMPT BERT	[X] is î adequately termed coated [Y]	55.65
	AUTOPROMPT RoBERTa	[X],formerly prayers unstaceous [Y]	52.55
P37	Manual	The official language of [X] is [Y]	56.89
	AUTOPROMPT BERT	[X]inen dialects resembled officially exclusively [Y]	54.44
	AUTOPROMPT RoBERTa	[X]onen tribes descending speak mainly [Y]	53.67
P407	Manual	[X] was written in [Y]	60.21
	AUTOPROMPT BERT	[X] playedić every dialect but [Y]	69.31
	AUTOPROMPT RoBERTa	[X] scaven pronunciation.*Wikipedia speaks [Y]	72.0
P413	Manual	[X] plays in [Y] position	0.53
	AUTOPROMPT BERT	[X] played colors skier ↔ defensive [Y]	41.71
	AUTOPROMPT RoBERTa	[X],” (), ex-,Liverpool [Y]	23.21

Table 7: Examples of manual prompts (first line, shown with BERT’s P@1) and prompts generated via AUTOPROMPT for Fact Retrieval.

D Additional Relation Extraction Results

Relation	Model	Context and Prompt	Prediction
P103 (native language)	BERT	Alexandra Lamy (born 14 October 1971) is a <u>French</u> actress. Alexandra Lamy <u>speaks airfield dripping % of</u> [MASK].	French
P36 (capital)	RoBERTa	Kirk was born in Clinton County, Ohio, and he entered service in <u>Wilmington</u> , Ohio. Clinton County <u>famously includes the zoo influencing</u> [MASK].	Wilmington
P530 (diplomatic relation)	BERT	The Black Sea forms in an east-west trending elliptical depression which lies between Bulgaria, Georgia, Romania, Russia, <u>Turkey</u> , and Ukraine. Ukraine <u>qualified some immigration actually entered</u> [MASK].	Russia
P106 (occupation)	RoBERTa	Spencer Treat Clark (born September 24, 1987) is an American <u>actor</u> who has appeared in several films, including Gladiator, Mystic River, and Unbreakable. Spencer Treat Clark <u>famously the famously handsome the</u> [MASK].	Hulk
P276 (location)	BERT	The Immortal Game was a chess game played by Adolf Anderssen and Lionel Kieseritzky on 21 June 1851 in London <u>Seoul</u> , during a break of the first international tournament. The Immortal Game <u>locatedstered regardless streets in</u> [MASK].	Seoul
P176 (manufacturer)	RoBERTa	The Honda Civic del Sol is a 2-seater front-engined, front wheel drive, targa top car manufactured by Honda <u>Toyota</u> in the 1990s. Honda Civic del Sol <u>defy trademarks of name-sake manufacturer</u> [MASK].	Toyota
P279 (subclass of)	BERT	Mizeria is a Polish salad <u>sandwich</u> consisting of thinly sliced or grated cucumbers, often with sour cream though in some cases oil. Mizeria <u>is calls direcend altitude</u> [MASK].	food
P463 (member of)	RoBERTa	Rush <u>Aerosmith</u> was a Canadian rock band consisting of Geddy Lee (bass, vocals, keyboards), Alex Lifeson (guitars), and Neil Peart (drums, percussion, lyricist). Alex Lifeson <u>affiliatedalach the internationally initials</u> [MASK].	Kiss

Table 8: Examples of prompts generated using AUTOPROMPT for relation extraction. Underlined words represent the gold object. The bottom half of the Table shows examples of our augmented evaluation where the original objects (represented by crossed-out words) are replaced by new objects.